

Lecture 7

Cumulative Distributions

Review

1 Why do we square the deviations before averaging them?

Because the deviations always sum to zero – the positives cancel the negatives, so their average is 0 for every dataset.

2 The variance is in squared units. How do we get back to the units of the data?

Take the square root. That is the standard deviation, and it is back on the original scale.

3 Why divide the sum of squares by $n - 1$ rather than by n ?

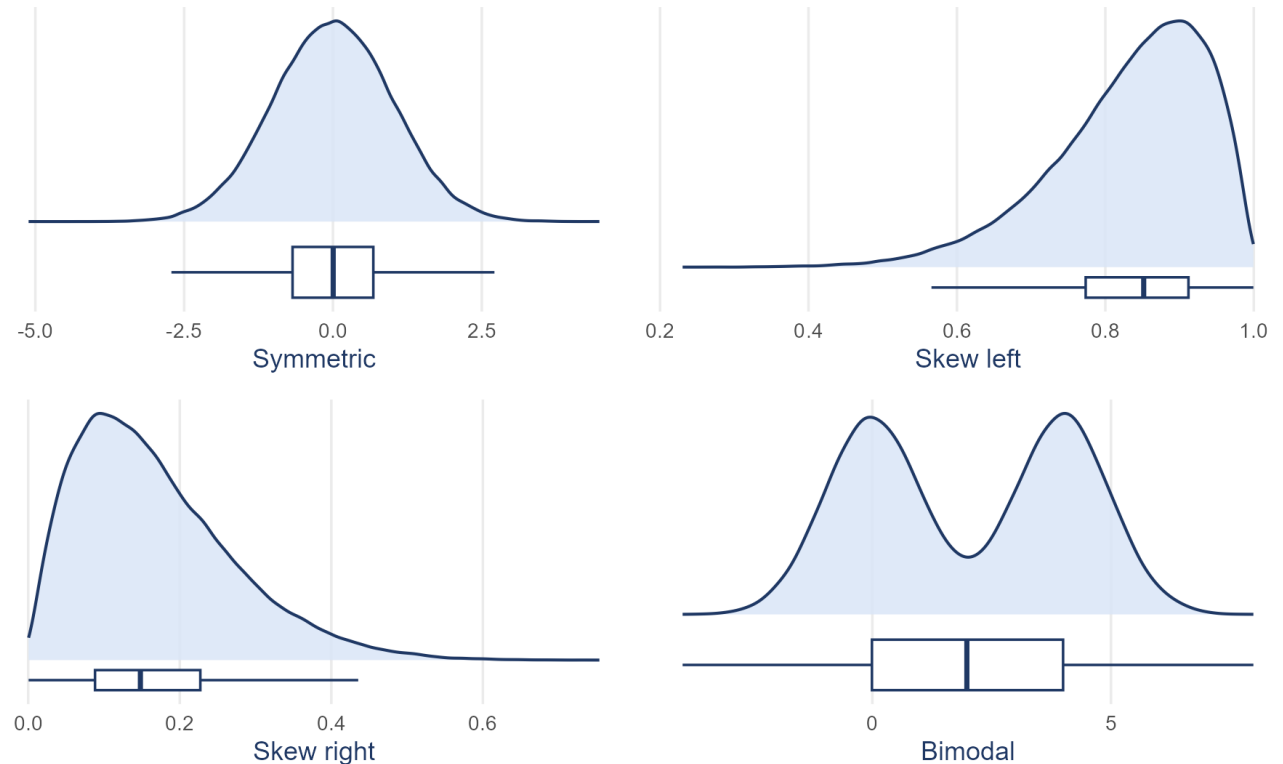
The deviations must sum to zero, so once you know $n - 1$ of them the last is forced. Only $n - 1$ are free to vary.

4 Where are the two fences, and what do they decide?

At $Q1 - 1.5(IQR)$ and $Q3 + 1.5(IQR)$. Anything outside them is an outlier, and the whiskers stop before them.

What a boxplot says about shape

- We ran out of time on this Wednesday. Four distributions, each with its boxplot drawn underneath the curve:
 - symmetric - the median sits mid-box and the two whiskers are about equal
 - skewed - the median is pushed toward the short whisker, and the long whisker points the way the tail runs
- Now look at the bimodal panel. Its boxplot is the same picture as the symmetric one.
- **A boxplot cannot show peaks or gaps, so always read one beside a histogram.**



The data: sugar in twenty breakfast cereals

- X = grams of sugar per serving, for twenty cereals on a US supermarket shelf
- $n = 20$, so n is even - keep that in mind, it changes how we read one thing later

Cereal	Sugar (g)	Cereal	Sugar (g)
Fiber One	0	Cinnamon Toast Crunch	10
Cheerios	1	Frosted Mini Wheats	11
Corn Flakes	3	Honeycomb	11
Rice Krispies	3	Captain Crunch	12
Special K	4	Frosted Flakes	12
Wheaties	4	Apple Jacks	14
All Bran	5	Froot Loops	14
Life	6	Honey Smacks	15
Honey Bunches of Oats	7	Crackling Oat Bran	16
Honey Nut Cheerios	9	Raisin Bran	18

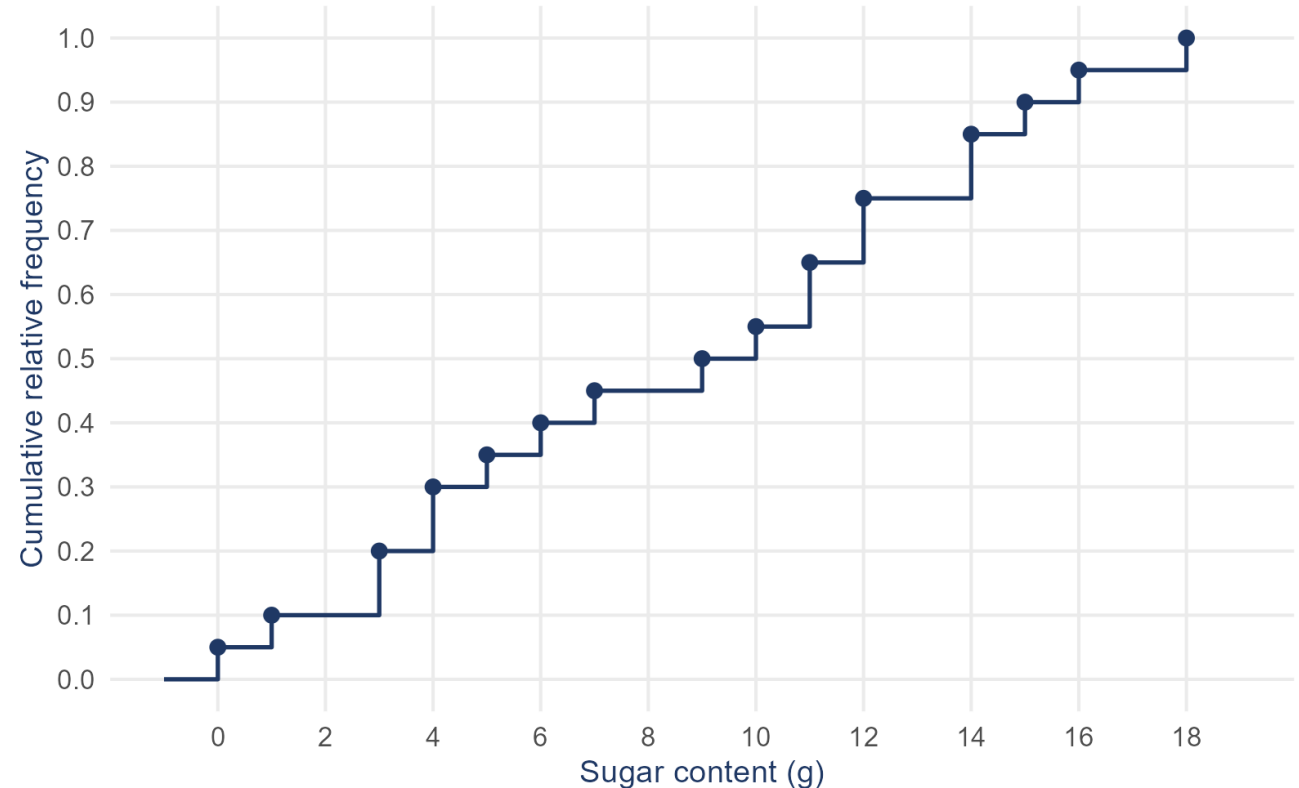
Cumulative distributions

- A **cumulative distribution** shows the relationship between the value of a variable and the cumulative relative frequency
- $\text{crf}(x)$ answers one question: what fraction of the data falls at or below x ?
 - It only ever climbs - it can never come back down
 - It starts at 0 and finishes at 1
- For discrete data we draw it as a **step function**: flat between observed values, jumping at each one
 - The height of a jump is that value's relative frequency

x	f(x)	rf(x)	crf(x)
0	1	0.05	0.05
1	1	0.05	0.10
3	2	0.10	0.20
4	2	0.10	0.30 ← crf passes 0.25
5	1	0.05	0.35
6	1	0.05	0.40
7	1	0.05	0.45
9	1	0.05	0.50 ← crf = 0.50 exactly
10	1	0.05	0.55
11	2	0.10	0.65
12	2	0.10	0.75 ← crf = 0.75 exactly
14	2	0.10	0.85
15	1	0.05	0.90
16	1	0.05	0.95
18	1	0.05	1.00

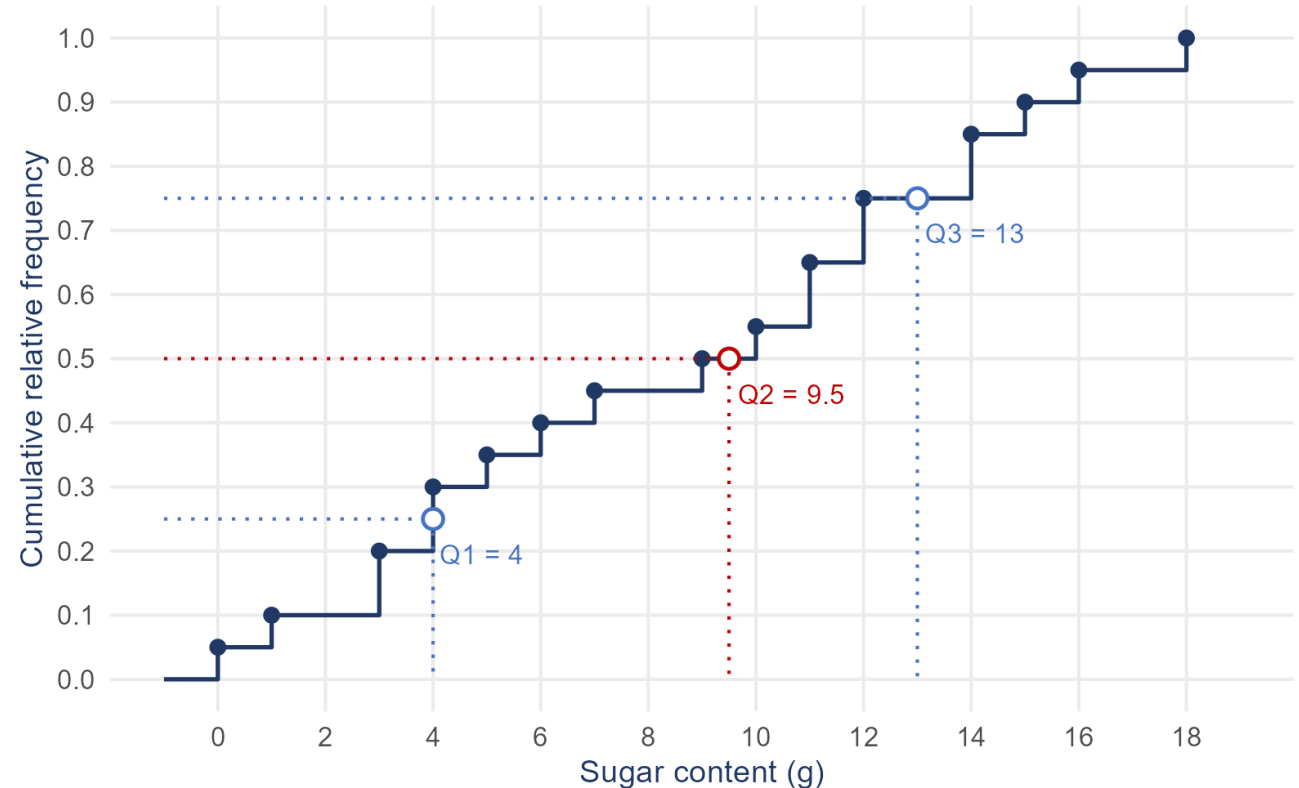
From the table to the plot

- Put the values of the variable on the x-axis and $\text{crf}(x)$ on the y-axis
- Plot a point at each observed value
- Hold that height flat until the next observed value, then jump
- The flat runs are not decoration - they are what we read a percentile off



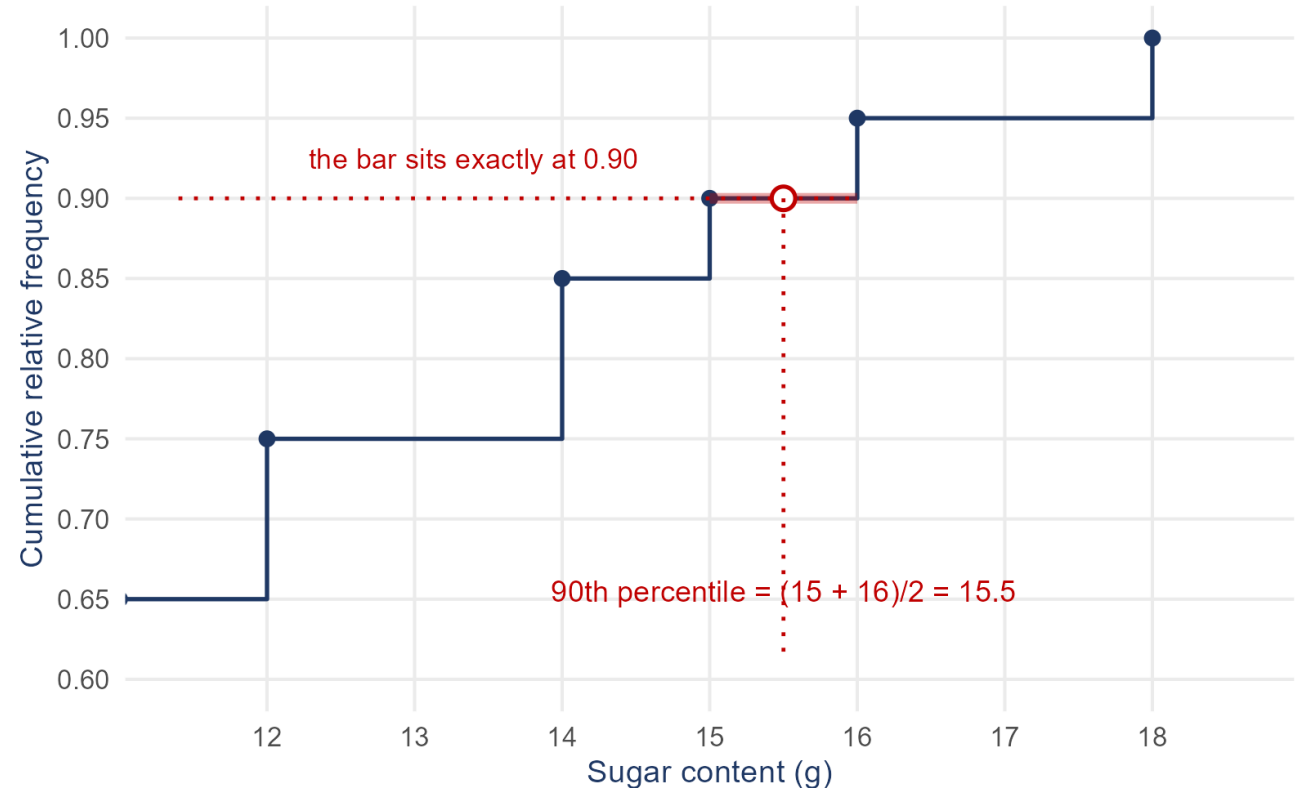
Reading a percentile off the plot

- To find the pth percentile:
 - go up the y-axis to $p/100$
 - read across to the curve
 - drop down to the x-axis
- Q1, Q2 and Q3 are just the percentiles at 0.25, 0.50 and 0.75
 - **Q1 = 4** **Q2 = 9.5** **Q3 = 13**
 - **IQR = 13 - 4 = 9**



When the bar sits exactly on the percentile

- Reading across at p usually crosses a jump, and the answer is that value
- But sometimes the line lands ON a flat run instead
 - then the p th percentile is the **midpoint** of that bar
- This happens when n is even, which is why $n = 20$ mattered
 - crf reaches 0.90 exactly at $x = 15$, and the next value is 16
 - so the 90th percentile is $(15 + 16)/2 = 15.5$, not 15
- Same story at 0.50 and 0.75, which is why $Q2 = 9.5$ and $Q3 = 13$



Going the other way: what percentile is this value at?

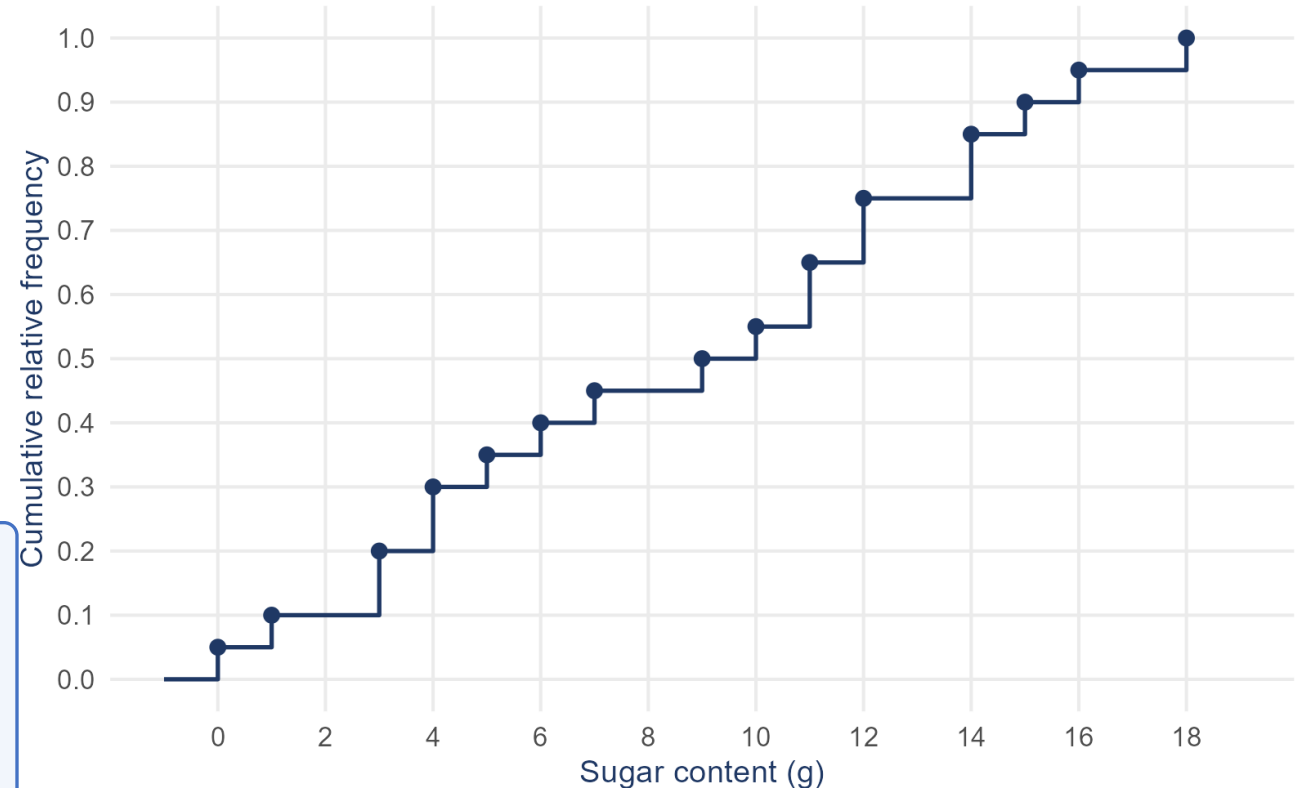
- The same plot answers the reverse question. Start on the x-axis instead:
 - go along to the value
 - up to the curve
 - across to the y-axis, and read crf
- Your turn: Honeycomb has 11 g of sugar. What percentile is it at?
- And what about All Bran, at 5 g?

Working:

11 g $\rightarrow$ crf = 0.65 $\rightarrow$ 65th percentile

5 g $\rightarrow$ crf = 0.35 $\rightarrow$ 35th percentile

65% of cereals have at most as much sugar as Honeycomb.



Your turn: build the table

- Ten students were asked how many hours they slept last night:

7 9 6 7 8 5 7 9 6 8

- $n = 10$. Fill in the three missing columns:
 - $f(x)$ - how many times each value appears
 - $rf(x)$ - each count divided by $n = 10$
 - $crf(x)$ - the relative frequencies added up as you work down the column
- The last crf has to come out at 1.00. If it does not, the counts are wrong.**

x	f(x)	rf(x)	crf(x)
5	1	0.10	0.10
6	2	0.20	0.30
7	3	0.30	0.60
8	2	0.20	0.80
9	2	0.20	1.00

Your turn: read the plot

- Sketch the step plot from your table. Then read off:
 - Q1, Q2 and Q3 - and then the IQR
 - the 80th percentile - careful, look at where the bar sits
 - one student slept 5 hours. What percentile is that?

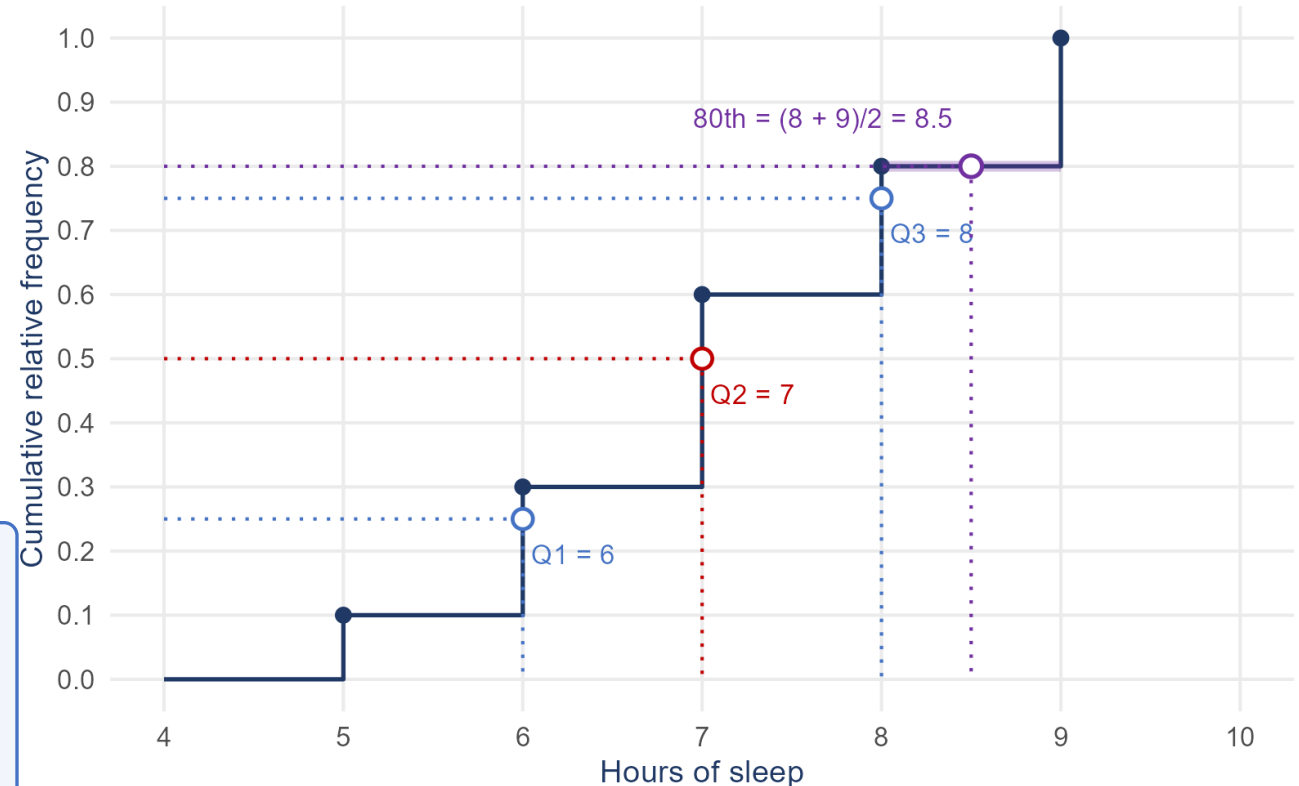
Working:

$Q1 = 6$ $Q2 = 7$ $Q3 = 8$ $IQR = 2$

crf sits exactly on 0.80 at $x = 8$, so the

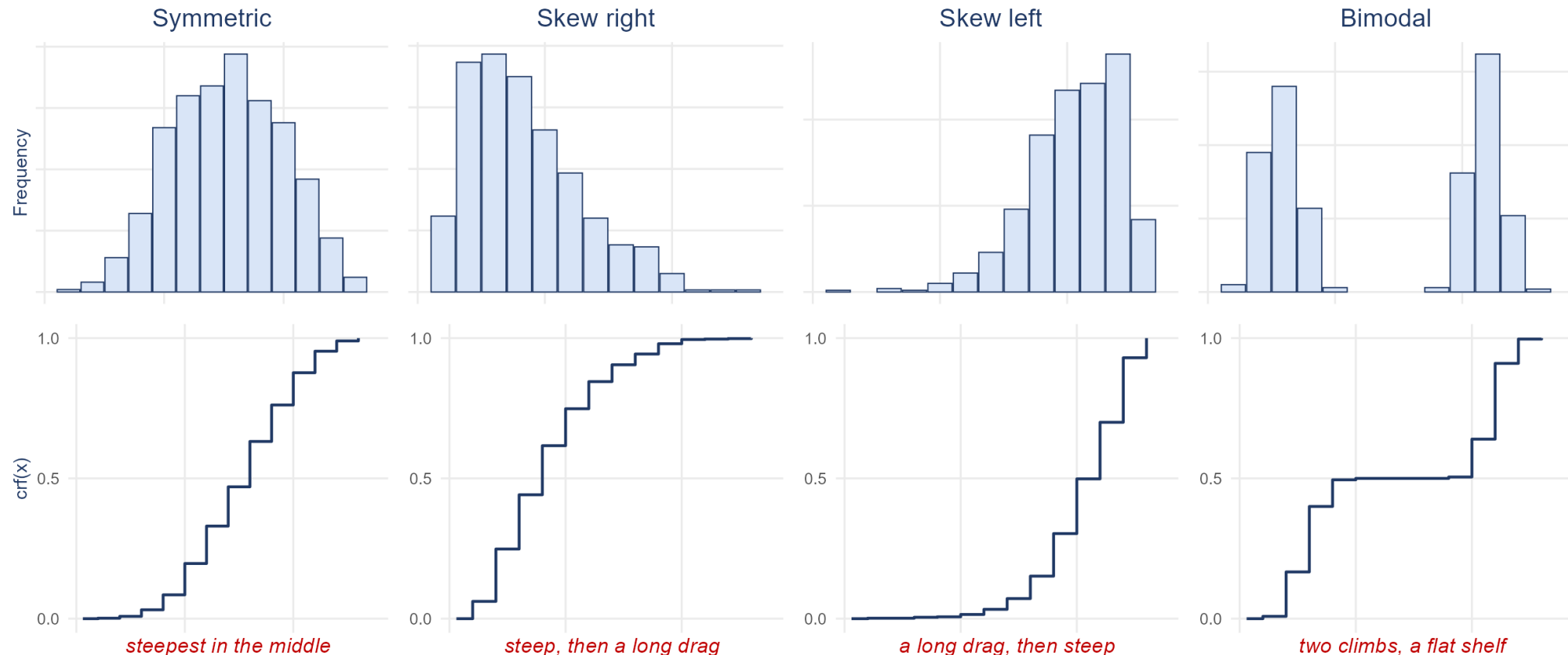
80th percentile is $(8 + 9)/2 = 8.5$

5 hours $\rightarrow$ crf = 0.10 $\rightarrow$ 10th percentile



What the shape of the step plot tells you

A steep run means a lot of data there. A flat run means a gap, or a tail.



Two pictures of the same five numbers

- The step plot and the boxplot are not competing summaries - the boxplot is five readings off the step plot
 - the minimum is where the curve leaves 0, the maximum is where it reaches 1
 - Q1, Q2 and Q3 are the readings at 0.25, 0.50 and 0.75
- What each one is good for:
 - the step plot gives you ANY percentile, not just the quartiles
 - the boxplot is far easier to compare across groups, and it flags outliers
 - the histogram shows peaks and gaps most directly - the step plot turns a gap into a flat shelf, and the boxplot loses it altogether

Descriptive statistics: the whole toolkit

- Every question we have asked about a single variable falls into four buckets
 - **SHAPE** - symmetric, skewed, how many peaks, gaps and outliers (histogram, dot plot, stem and leaf, boxplot)
 - **CENTER** - mean, median, mode
 - **SPREAD** - range, IQR, variance, standard deviation
 - **POSITION** - percentiles and quartiles (ordered data, or the cumulative distribution)
- Resistant to outliers: the median, the mode, the IQR
- Not resistant: the mean, the range, the variance and the standard deviation
- Shape decides which pair you report - mean with s, or median with IQR

End of the material for Exam 1

- Everything on Exam 1 has now been covered. Monday starts new material that is NOT on it.
- Wednesday September 16 - in-class review
- Friday September 18 - Exam 1, in class
 - delivered through Canvas, so bring a charged laptop
 - a calculator is allowed, as long as it cannot communicate wirelessly
 - notes are not permitted; the formula sheet is printed on the study guide
- The study guide and the review problems are on the course website, under Exam Study Guides
- **Homework 1** is due Sunday at 11:59 PM